

Grady Olsen

UCLA Agentic AI Course

Assignment 1 - Create a research assistant using Crew AI

Design

The structure of the agents was already pretty clearly spelled out in the assignment. There are four agents, one of which uses a single tool, and four tasks defined in order to complete a research project. The design choice I struggled with at first was where to put the most detailed design guidance: the agents or the tasks. I opted to keep the agent goal and backstory a bit more high level and flexible and put the very detailed instructions in the task description and expected output. This seemed to me like a good practice so that these agents could be re-used in situations with similar needs but other tasks could be swapped in if, for example, a different style of output was needed.

ReAct

Honestly, I see little to no evidence of ReAct behavior in my trace files. I even ran it through Claude to make sure I wasn't missing something and Claude agreed. Perhaps it is because I did not instruct any of the agents to iterate specifically, but I was expecting to see some evidence of observation and making some adjustments. With more time and in the upcoming weeks I will look to understand this mechanism better.

What Worked and What Did Not

Overall, the crew ran successfully and produced very good results. I did have some debugging to do with model access and then some silly errors where I did not comment out some lines that I should have. But once it ran it performed well.

The one exception is that on the first run the agents cited content almost exclusively from a single source. I found that problematic so I edited the search_task to specifically tell it to run each query separately and also to make sure that it was gathering results from many sources, not just the top search result. After making those changes my final report cited 5-6 sources.

What is Needed for These Agents to be Used on Production

I don't have a strong sense yet of how to answer this question but I have a couple of thoughts. One would be to improve the error handling to make it easier to resolve any issues. Also, the verbose trace log is helpful but it is a lot of text and some of it not necessarily useful. I think the agents could be improved upon to make it easier to see and understand what is happening at each step and what adjustments or tuning could be done to improve results. I might want to produce a second, nicely formatted output that is an explanation of how the report was created.